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Effect of a blend of palmitic and stearic acid on transfer of preformed fatty acids into milk fat using an 18 C n-3 fatty acid perturbation model

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ABSTRACT

The transfer of dietary fatty acids (FA) into milk fat is variable and has been reported to be higher on lower fat diets, which may be a homeorhetic adaptation to support milk fat production. The objective of this study was to determine the effect of increasing dietary fat on the kinetics of transfer of preformed FA into milk fat using a perturbation model based on abomasal infusion of a bolus of n-3 FA. Eight ruminally cannulated multiparous Holstein cows (115 ± 32 DIM and producing 41.5 ± 4.43 kg of milk; mean $\pm$ SD) were used in a switchback design with three 21-d periods. Treatments were a low-fat basal diet (CON) and the basal diet supplemented with a blend of palmitic and stearic FA fed at 2% DM (Fat; 33.5% 16:0, 53.1% 18:0, and 6.2% 18:1n-9). Increasing dietary FA tended to increase milk yield by 2.2 kg/d and increased milk fat yield by 170 g/d. Yield of preformed FA in milk was increased 119 g/d but yield of <16 C and 16 C FA was not changed compared with CON. Apparent total-tract digestibility of DM, OM, NDF, and FA were not changed by FA supplementation. Transfer of dietary 18 C FA to milk fat was 110% of dietary fat consumed in CON and was decreased to 82% in the Fat treatment. On d 15 of each period, cows were abomasally infused with a free FA stock that provided 44 g of α -linolenic acid (ALA) and 20 g of stearidonic acid (SDA), and milk FA profile was observed over the following 7 d. Maximal enrichment of ALA and SDA in milk fat occurred at the second milking following the bolus (~14 h) and returned to baseline concentrations by 74 h. Increasing dietary fat decreased total transfer of n-3 FA by 5.4 percentage units. Total transfer of infused ALA was 51.9% and SDA was 27.9% in the CON and decreased to 43.3% and 22.7% in the high-fat diet, respectively. Similar to previous work, the majority (81%) of the n-3 FA transferred to milk fat occurred in the first 24 h, which is expected to represent

direct transfer of absorbed FA to milk fat and was not influenced by dietary fat. The greater than 100% transfer efficiency of 18 C FA to milk fat and the higher transfer efficiency of n-3 FA to milk fat when feeding a lower fat diet indicates dynamic partitioning of absorbed FA that is responsive to dietary FA supply.

Key words: omega-3, milk fat yield, lipid metabolism

INTRODUCTION

Market demand for milk fat has driven an increase in both the genetic potential for milk fat yield in dairy cows in the United States and a focus on nutritional management to support its synthesis. Milk fatty acids (FA) originate either from de novo synthesis in the mammary gland or uptake from the plasma, and dietary fat is important to supporting milk fat yield. At the same time, there are major limitations on feeding FA to dairy cows because UFA disrupt rumen fermentation, the cow has a limited capacity for FA digestion, and FA are an expensive nutrient. The meta-regression of Khiaosa-ard et al. (2015) observed that transfer efficiency of 18 C FA to milk fat is highly variable, is often over 100%, and is decreased with increasing dietary fat. Low FA intake relative to demand for milk fat synthesis may induce homeorhetic adaptations that prioritize mammary uptake of absorbed FA and increase transfer efficiency by partitioning FA to milk fat. However, regulation of trafficking of absorbed FA and the impact of dietary fat level has not been well investigated.

Work using ^{14}C isotope approach over 40 yr ago established that over 85% of the preformed FA came directly from dietary absorption and were transferred in a fast pool during the first 24 h, whereas the remaining 15% came from cycling of FA through other body tissues after 24 h (Palmquist and Mattos, 1978). The mammary gland has little to no capacity to synthesize 18 C FA, allowing a direct observation of their uptake by the mammary gland and abomasal infusion studies have reported a rather consistent ~55% transfer of duodenal infused 18 C FA into milk fat (Enjalbert et al., 1998). However, recent work with increasing dietary 18 C FA through feeding

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whole cottonseed or high oleic soybeans observed a reduction in transfer efficiency of 18 C FA as dietary fat increased (Bales and Lock, 2024; Pierce et al., 2024). Understanding how dietary fat affects the utilization of preformed FA is expected to provide critical insight into the regulation of mammary FA absorption.

Investigating FA trafficking in dairy cows is physiologically and technically challenging. Ideally, stable ^{13}C labeled FA would be used as a tracer (Mosley et al., 2006). However, they are prohibitively expensive to use, especially at the doses needed for large animals with high metabolic rates. Urrutia et al. (2023) established a perturbation model to study kinetics of FA trafficking in dairy cows based on abomasal infusion of n-3 FA. The small metabolic pools and low concentration of n-3 FA in milk fat allow the efficient use of a moderate bolus (79.4 g) of n-3 FA without super-physiological doses. Urrutia et al. observed that transfer of n-3 FA to milk fat fit a 2-pool model with rapid transfer during the first ~24 h, which is expected to represent direct transfer through chylomicrons, and slow transfer from 24 to 102 h, representing indirect transfer through tissue recycling. They observed that α -linolenic acid (ALA; 18:3n-3) followed similar transfer rates and kinetics to ^{14}C palmitic acid described by Palmquist and Conrad (1971). This provides the opportunity to use a bolus of n-3 FA as a model to investigate partitioning of absorbed FA to milk fat.

Preformed FA are a main contributor to milk fat synthesis and a better understanding of their partitioning to milk fat is needed to develop optimal FA feeding recommendations. Our objective was to determine the effect of increasing dietary FA with a blend of palmitic and stearic acid on the partitioning of absorbed FA to milk fat. We hypothesize that increasing dietary FA would decrease the rate of transfer of absorbed FA to milk fat observed both as total dietary 18 C transferred to milk fat and the transfer of an abomasally infused bolus of n-3 FA.

MATERIALS AND METHODS

The experiment and animal-related procedures were approved by the Pennsylvania State University Institutional Animal Care and Use Committee (PROTO202102054). The experiment was conducted from March to June of 2024 at Penn State Dairy Research Center in University Park, Pennsylvania.

Animals, Diets, and Experimental Design

Eight multiparous Holstein cows permanently fitted with rumen cannulas (110 mm; <https://rumencannula.com/>) were randomly assigned to treatment sequence in a switchback design with 21-d experimental periods. There was a 14-d pretrial period to allow adaptation to the ties-

tall barn and the experimental basal diet. The cows were 115 ± 32 DIM and producing 41.5 ± 4.43 kg/d of milk that contained $4.34 \pm 0.50\%$ milk fat and $3.34 \pm 0.35\%$ milk protein at the end of the pretrial period (mean $\pm$ SD). Treatments were a low-fat basal diet with no higher-fat byproduct, oilseeds, or fat supplements (CON) and the basal diet with the addition of a FA supplement that predominantly contained a blend of 16:0 and 18:0 and 6.16% 18:1n-9 at 2.0% DM (Fat; Endense-V, Perdue Agribusiness, Salisbury, MD; Tables 1 and 2). The basal low-fat diet was mixed in a stationary mixer (Electra-Mix 1062, I.H. Rissler Manufacturing, Mohnton, PA) and the FA prill was mixed into the basal diet in a mobile mixer (Rissler 1050 Feedcart, I.H. Rissler Manufacturing). Cows were fed once daily at ~0700 h for 5% refusals. Dry matter of diet ingredients was calculated weekly (55°C forced-air oven for 48 h) and diets were adjusted accordingly.

Preparation of n-3 Free FA and Bolus Infusion

A free FA stock high in n-3 FA was prepared by saponification of Ahiflower oil (Omega Match, Purina) using ethanolic KOH followed by acidification to pH 4 with HCl in water (1:1) and hexane extraction (Urrutia et al., 2018). On d 15 of each experimental period, 120 mL of the free FA stock containing 44 g of ALA (18:3n-3) and 20 g of stearidonic acid (SDA; 18:4n-3) were abomasally infused in all cows. The infusate was delivered in 3 hourly 40-mL infusions (3.5, 2.5, and 1.5 h before the 1800 h milking).

Abomasal infusion lines were placed the day before each bolus by passing tubing with a flange on the end through the reticulo-rumen orifice as described by Spires et al. (1975). The line was rinsed with ~20 mL of water before the infusions and with ~10 mL of 70% ethanol and ~10 mL water following the bolus.

Feed and Feces Sampling and Analysis

Feed intake of each cow was measured daily from d 17 to 21. On d 17 through 21, a representative sample of each feed ingredient and one-eighth of refused feed from each cow was collected, composited within cow within period for refusals and period for feeds, and stored at -20°C for determination of chemical composition. Feces were collected by grab sampling on d 17 through 21 at 1200, 2000, 0400, 1600, 0000, and 0800 h; composited by cow within period; and stored at -20°C .

All samples were lyophilized (Ultra 35-XL, Virtis Co. Inc., Gardiner, NY) and ground through a 1-mm screen in a Wiley mill (A.H. Thomas Co., Philadelphia, PA). Feed ingredients were analyzed for 105°DM (method 930.15; AOAC International, 2000), starch (Hall et al., 2015),

Table 1. Ingredients and nutrient composition of experimental diets

Item	Treatment ¹	
	CON	Fat
Ingredient, % DM		
Corn silage ²	49.2	48.2
Alfalfa haylage ³	13.2	12.9
Canola meal	10.9	10.7
Corn gluten feed	6.91	6.77
Molasses	5.10	4.99
Grass hay ⁴	3.74	3.66
Ground corn	3.27	3.21
Corn gluten meal	2.49	2.44
FA supplement ⁵	—	2.07
Vitamin-mineral ⁶	1.79	1.75
Sodium bicarbonate	1.00	0.98
DDGS ⁷	0.93	0.91
Blood meal	0.54	0.53
Lysine ⁸	0.48	0.47
Urea ⁹	0.37	0.36
Nutrient composition, % DM		
NDF	31.9	31.3
NDFom	30.8	30.2
Starch	26.6	26.1
ADF	18.8	18.4
CP	16.0	15.7
iNDF	10.4	10.2
Ash	5.72	5.60
Total FA	2.66	4.66
16:0	0.44	1.12
18:0	0.08	1.16
18:1n-9	0.62	0.73
18:2n-6	1.18	1.18
18:3n-3	0.17	0.17
16 C	0.45	1.13
18 C	2.14	3.42

¹Dietary treatments were a low-fat basal diet (CON) and a high-fat diet that fed a fatty acid (FA) supplement that contained a blend of stearic and palmitic acid at 2% diet DM (Fat).

²Corn silage contained 32.6% NDF, 9.87% iNDF, 18.7% ADF, 6.5% CP, 46.0% starch, 2.89% ash, and 2.63% FA (0.25% 14:0, 15.7% 16:0, 0.14% 16:1, 2.50% 18:0, 22.9% 18:1 *cis*-9, 51.4% 18:2n-6, 2.03% 18:3n-3) on a DM basis.

³Alfalfa haylage contained 44.2% NDF, 23.6% iNDF, 35.7% ADF, 20.0% CP, 0.83% starch, 8.51% ash, and 1.70% FA on a DM basis.

⁴Grass hay-straw contained 63.2% NDF, 18.1% iNDF, 35.7% ADF, 10.0% CP, 0.7% starch, 6.2% ash, and 1.16% FA on a DM basis.

⁵Fatty acid prill contained 33.5% 16:0, 53.1% 18:0, and 6.16% 18:1 *cis*-9 acid (Endense-V, Perdue Agribusiness, Salisbury, MD).

⁶Vitamin-mineral mix contained (as-fed basis) 37.03% calcium carbonate, 29.90% distillers grain, 24.90% salt, 4.20% magnesium oxide, 2.42% monocalcium phosphate, and 1.95% trace mineral. Composition (DM basis): 13.34% Ca, 0.74% P, 2.60% Mg, 0.33% K, 10.39% Na, 0.52% S, 652 mg/kg Cu, 1,718 mg/kg Zn, 783 mg/kg Fe, 13.00% mg/kg Se, 88,958 IU/lb vitamin A, 28,400 IU/lb vitamin D, and 850 IU/lb vitamin E.

⁷DDGS = dried corn distillers grains with solubles.

⁸Rumen-protected lysine (AjiPro-L, Ajinomoto Co. Inc.).

⁹Coated urea fed as Optigen (256% CP on a DM basis; Alltech Inc.).

CP (method 990.3; AOAC International, 2000), ADF (method 973.18; AOAC International, 2000), NDF with amylase on an OM basis (Van Soest et al., 1991), and in vitro 240-h indigestible NDF (iNDF) by Cumberland Valley Analytical Services (Forage Lab, Waynesboro,

Table 2. Fatty acid (FA) profile of the dietary supplement and n-3 free FA stock used in the bolus infusion

Item	Prill ¹	n-3 bolus ²
FA, % of FA		
14:0	0.84	0.03
16:0	33.5	5.93
16:1 <i>cis</i> -9	0.01	0.09
17:0	0.15	0.05
18:0	53.1	2.09
18:1 <i>trans</i> ³	3.37	ND ⁴
18:1 <i>cis</i> -9	6.16	9.41
18:1 <i>cis</i> -11	0.19	0.59
18:2n-6	0.81	11.8
18:3n-6	0.88	5.38
20:0	0.45	0.06
18:3n-3	0.01	44.4
18:4n-3	—	18.1
Others ⁵	2.47	2.21
Total, % DM	99.3	97.0

¹Prilled FA supplement containing a blend of palmitic and stearic acid (Endense-V, Perdue Agribusiness, Salisbury, MD).

²Free FA stock derived from Ahiflower oil (Omega Match, Purina) that delivered 44 g of ALA and 20 g of SDA in the bolus.

³Sum of all *trans* 18:1 FA.

⁴ND = below the level of detection.

⁵Includes FA >20 C.

PA). Refusals were analyzed for DM, ash, and NDF and feces were analyzed for DM, ash, NDF, and iNDF by the same methods. Total FA concentration and profile of feeds, refusals, and fecal samples were determined by direct methylation in 10% methanolic HCl as described by Sukhija and Palmquist (1988), followed by quantification by GC with a flame ionization detector as described by Rico et al. (2014). The n-3 free FA stock used in the bolus infusion was methylated in 0.5 M sodium methoxide in methanol followed by 10% methanolic HCl for FA analysis as described by Jenkins (2010) and profile analyzed as stated previously. Peaks were identified using methyl ester standards (GLC 461, 566, and 780; Nu-Chek Prep Inc., Elysian, MN), and 18:4n-3 was identified based on retention order and previous experience.

Milk Yield and Composition

Milk yield at each milking (0630 and 1800 h) was automatically recorded by an integrated milk meter (DeLaval milk meter, MM27BC, DeLaval, Gurnee, IL). Milk weights were adjusted based on stall deviations calculated by modeling the effect of milking (a.m./p.m.), day, cow, and parlor stall using data from the entire herd over the 7 d before each observation period (Andreen et al., 2020). Milk was sampled at each milking on d 14 to 21 for determination of fat, protein, and urea-N by mid-infrared spectroscopy (LactoScope FT-A, PerkinElmer Inc., Shelton, CT). Milk SCC was measured on the last day each period by flow cytometry (DairyOne DHIA,

Ithaca, NY). Milk FA profile was determined at each milking on d 14 to 16 and on a daily composite on d 17 to 21. Briefly, milk FA were extracted from whole milk using hexane-isopropanol and transmethylated using sodium methoxide as described by Rico and Harvatine (2013) and FA profile determined by GC as described by Baldin et al. (2018) with standards as described previously.

Calculations

Indigestible NDF was used as an internal flow marker to calculate apparent total-tract (TT) digestibility of DM, OM, NDF, and FA using the marker-ratio method as described by Huhtanen et al. (1994). Nutrient intake was calculated as the difference of offered and refused feed across d 17 to 21.

Yield of individual FA in milk fat was calculated as described by Glasser et al. (2007), and transfer of dietary FA into milk was calculated both as a percentage of consumed FA and absorbed FA across d 17 to 21 of each period.

Transfer of n-3 PUFA into milk following the bolus was determined by subtracting from the pre-bolus concentration on d 14 (Urrutia et al., 2018, 2023). Samples analyzed at each milking were subtracted from the same milking time before the bolus and samples composited within day were subtracted from the weighted average of the day before the bolus. Total transfer was calculated by integration over the first 158 h, as milk ALA and SDA concentrations returned to pre-infusion concentrations by 74 h. Transfer over the first 24 h and from 24 to 158 h were also calculated based on the fast and slow transfer pools described by Palmquist and Conrad (1980) and Urrutia et al. (2023). Transfer kinetics were not observed in one cow in period 2 due to unexpected blockage of the abomasal infusion line at the time of the bolus.

Statistical Analysis

Milk yield and milk component records from d 17 to 21 of each period were averaged for each cow in each period and analyzed using the REML method of JMP Pro (version 17.0; SAS Institute Inc., Cary, NC) in the following mixed model:

$$Y_{ijk} = \mu + C_i + P_j + T_k + \varepsilon_{ijk},$$

where Y_{ijk} represents the response variable of interest, μ is the overall mean, C_i is the random effect of cow ($i = 1$ to 8), P_j is the random effect of period ($j = 1$ to 3), T_k is the fixed effect of treatment ($k = 1$ to 2), and ε_{ijk} is the residual error. The pretrial observation was also included

as a covariate for DMI, milk yield, milk components, and milk FA profile.

Transfer of n-3 PUFA into milk over time was analyzed using the repeated statement of PROC MIXED in SAS (version 9.4; SAS Institute Inc.) using the pre-bolus d 14 observation as a covariate. Fixed effects were treatment, time, and their interaction and random effects were cow and period. The compound symmetry covariate structure was used, time was the repeated variable, subject was cow, and denominator degrees of freedom were adjusted by the Kenward–Roger method.

In all analyzes, studentized residuals outside of ± 3.0 were considered outliers and removed. Significance for main effects and interaction was declared at $P \leq 0.05$ and $P \leq 0.10$ and tendencies between 0.05 and 0.10 and 0.10 and 0.15, respectively.

RESULTS AND DISCUSSION

This study was designed to understand the effect of increasing dietary FA concentration on the transfer of dietary FA to milk fat. The meta-analysis of Khiaosaard et al. (2015) observed >100% transfer efficiency of dietary 18 C FA to milk when cows were fed lower fat diets, so the control was designed to have minimal fat. A blend of palmitic and stearic acid was selected as the fat supplement to minimize confounding effects on rumen fermentation and possible bioactive FA that reduce milk fat (Table 1). The fat supplement was fed at 2% of diet DM to adequately increase FA intake without overrunning digestive capacity. The diets were successful in maintaining normal milk fat yield and low milk *trans*-10 18:1 levels, and the supplement did not decrease FA digestibility as described subsequently. There also was no influence of the n-3 bolus on milk FA by source following the bolus infusion in the current experiment (no treatment by day interaction; Supplemental Figure S1, see Notes).

Milk Production

The FA supplement tended to increase milk yield 2.2 kg/d compared with CON (37.3 vs. 39.5 kg/d, $P = 0.07$). Milk fat concentration averaged 3.99% in the control and was not changed by treatment, but the high FA diet increased milk fat yield by 170 g/d (Table 3; $P < 0.02$). Milk fat concentration in the Northeast Milk Market averaged 4.00% during the period of the study, indicating the basal diet resulted in normal milk fat. Increasing dietary fat decreased milk protein by 0.06 percentage units but did not change protein yield or MUN.

Production responses to mixtures of palmitic and stearic acid have been well studied and are summarized in the meta-analysis of dos Santos Neto et al. (2021). Simi-

Table 3. Effect of a blend of palmitic and stearic acid on milk production and milk fatty acid (FA) concentration and yield in mid-lactation cows compared with a low-fat control

Item	Treatment ¹		SEM	P-value ²
	CON	Fat		
Yield, kg/d				
Milk	37.3	39.5	2.51	0.07
Fat	1.49	1.66	0.10	0.02
Protein	1.25	1.30	0.08	0.21
FA by source, ³ g/d				
<16 C	381	410	25.4	0.58
16 C	426	481	33.8	0.39
>16 C	452	571	41.0	0.01
Total 18 C	441	559	39.5	0.01
OBCFA ⁴	44.9	45.9	7.89	0.37
18:1 <i>trans</i> -10	8.32	9.63	1.10	0.08
Concentration, %				
Fat	3.99	4.20	0.12	0.23
Protein	3.31	3.25	0.06	0.03
FA by source, ³ % FA				
<16 C	28.4	26.5	1.11	0.03
16 C	31.5	31.0	0.82	0.17
>16 C	34.8	38.5	1.23	<0.01
Total 18 C	33.4	36.9	1.28	<0.01
OBCFA ⁴	3.33	3.02	0.55	<0.01
18:1 <i>trans</i> -10	0.62	0.60	0.07	0.70
MUN, mg/dL	10.5	9.88	1.44	0.24

¹Dietary treatments were a low-fat basal diet (CON) and a high-fat diet fed a FA supplement that contained a blend of stearic and palmitic acid at 2% diet DM (Fat). Samples represent the last 3 d of each period after the infused n-3 FA had been cleared.

²Effect of treatment.

³Fatty acids <16 C calculated as even straight-chain FA less than 16 C that originate from mammary synthesis, FA >16 C originated from plasma absorption, and 16 C FA is the sum of 16:0 and 16:1 that originate from both sources.

⁴OBCFA is the sum of odd- and branched-chain FA.

lar to the current work, these blends generally result in a small increase in milk yield (0.47 kg/d) but increase milk fat yield 23.4 g/d per 1 percentage unit inclusion. Thus, the milk fat yield response in the current study was much larger than most previous experiments, although responses are variable. For example, Burch et al. (2021) reported a 60 g/d increase in milk fat with a 60:30 mix of palmitic and stearic acid and Shepardson and Harvatine (2021) reported a 140 g/d increase in milk fat yield with a 45.3% palmitic and 49.1% stearic acid prill compared with a no-supplement control. The variation in response could be due to the heterogeneity of the basal diets, specifically the basal fat concentration, or interactions with the physiological potential of the cows.

Milk FA by Source

Milk FA originate from mammary synthesis and uptake from plasma. Increasing dietary fat commonly decreases the concentration and yield of <16 C FA that are synthesized in the mammary gland (Rabiee et al., 2012) and

likely the de novo portion of 16 C FA, although that is not determined in traditional experiments. This is commonly referred to as a “substitution” effect and theoretically allows the cow to maintain a desired level of milk fat yield while reducing the metabolic burden of de novo synthesis. In the current experiment, the concentration of <16 C FA in milk fat was decreased 1.9 percentage units but FA >16 C was increased by 3.2 percentage units with Fat ($P < 0.05$), while there was no change in the concentration of 16 C FA ($P = 0.17$; Table 3). However, due to the increase in total fat yield, the yield of <16 C de novo synthesized and 16 C mixed source FA did not change, whereas the yield of >16 C preformed FA was increased 119 g/d ($P = 0.01$) and total 18 C FA increased 118 g/d with the high-fat diet (Table 3), mostly due to increased yield of 18:0 and 18:1 *cis*-9 (Supplemental Table S2, see Notes).

The increase in milk preformed FA in the current experiment was larger than most other experiments feeding a blend of palmitic and stearic acid. The meta-analysis of dos Santos Neto et al. (2021) also reported that blends of 16:0 and 18:0 fed at $\leq 3\%$ of diet DM increased preformed FA by 42 g/d on average, with every 1 percentage unit increase in the supplement increasing milk preformed FA by 18 g/d. This would predict a 36 g/d increase for the current experiment. In specific experiments similar to the current treatment, de Souza et al. (2018) fed a near equal mix of 16:0 and 18:0 at 1.5% DM that did not change <16 C de novo and 16 C mixed FA but increased >16 C preformed FA by 32 g/d, which was not sufficient to increase total milk fat yield. However, in Shepardson and Harvatine (2021), a blend of 16:0 and 18:0 fed at 1.95% DM did not change <16 C FA yield but 16 C and >16 C FA were increased 91 and 62 g/d. The profile of FA supplemented in the current study is also similar to that fed by Western et al. (2020), but they only observed a 24 g/d increase in >16 C preformed FA, which may be due to decreased FA digestibility in their study.

Intake and Nutrient Digestibility

Increasing dietary FA did not change DMI but increased intake of total, 16 C, and 18 C FA by 522, 178, and 338 g/d, respectively (Table 4). The type and FA profile of fat supplements are important factors affecting DMI with UFA commonly decreasing intake (Drackley et al., 1992), whereas blends of 16:0 and 18:0 generally have no effect (Rabiee et al., 2012).

Apparent TT digestibility of DM, OM, and NDF were not changed by treatment ($P > 0.24$). With no change in DMI or digestibility, there was no effect of treatment on the total amount of DM, OM, and NDF digested ($P > 0.50$). Saturated FA supplements are expected to be rumen inert and not decrease fiber digestibility (Weld and

Table 4. Effect of a blend of palmitic and stearic acid on apparent total-tract digestibility of mid-lactation dairy cows compared with a low-fat control

Item	Treatment ¹		SEM	P-value ²
	CON	Fat		
Intake, kg/d				
DM	23.7	24.1	1.59	0.68
OM	22.5	23.1	1.47	0.58
NDF	7.33	7.38	0.58	0.87
FA	0.621	1.143	0.057	<0.001
16 C	0.103	0.278	0.012	<0.001
18 C	0.501	0.839	0.043	<0.001
Absorbed kg/d				
DM	14.8	15.3	1.07	0.51
OM	14.7	15.4	0.98	0.27
NDF	2.64	2.78	0.25	0.54
FA	0.369	0.701	0.038	<0.001
16 C	0.062	0.177	0.007	<0.001
18 C	0.322	0.532	0.031	<0.001
Digestibility, %				
DM	62.3	63.0	0.92	0.55
OM	64.9	65.9	0.59	0.18
NDF	36.1	39.2	2.34	0.24
FA	59.0	60.9	1.82	0.44
16 C	61.2	63.3	1.13	0.28
18 C	63.6	62.9	1.79	0.80
Transfer efficiency of 18 C FA, %				
Intake ³	69	53	11	<0.01
Absorbed ⁴	110	82	19	0.01

¹Dietary treatments were a low-fat basal diet (CON) and a high-fat diet that fed a FA supplement that contained a blend of stearic and palmitic acid at 2% diet DM (Fat). Samples represent the last 3 d of each period after the infused n-3 FA had been cleared.

²Effect of treatment.

³Calculated as milk FA as a percentage of FA intake.

⁴Calculated as milk FA as a percentage of FA absorbed.

Armentano, 2017). Although supplementing 18:0 and blends of 16:0 and 18:0 generally do not affect apparent TT NDF digestibility, Burch et al. (2021) reported a 3 percentage unit increase in NDF digestibility with a 60% 16:0 and 30% 18:0 blend. Highly enriched 16:0 supplements have been reported to increase NDF digestibility in vivo and in vitro (e.g., Sears et al., 2020; Western et al., 2020), but some experiments have observed no effect (Rico et al., 2014; Shepardson and Harvatine, 2021). The variation in response and mechanism of positive effects on NDF digestibility is not clear but may include providing FA for substrate for membrane synthesis, direct stimulation of microbial growth, or increased rumen residence time due to cholecystokinin secretion, which would be expected to decrease gut motility and increase rumen residence time (Choi et al., 2000).

There was also no effect of treatment on total, 16 C, and 18 C FA TT digestibility ($P > 0.28$). Increasing dietary fat generally is associated with a decrease in FA digestibility (Boerman et al., 2015). This is commonly interpreted as a limitation in FA emulsification and absorption but may also be due to differences between digestibility of fat sources that change as dietary fat is

increased. The meta-analysis by dos Santos Neto et al. (2021) reported a 3.11 percentage unit decrease in total FA digestibility per percentage unit increase of FA prills that were blends of 16:0 and 18:0. The FA supplement in the current experiment contained 6.16% 18:1 *cis*-9, which may have aided absorption. Oleic acid is an emulsifier and softens FA prills. The extent of rumen escape of 18:1 *cis*-9 in FA prill is not clear, but the portion that reaches the small intestine would be expected to increase emulsification of FA. It is interesting to note that lower fat diets and oilseeds normally have higher digestibility and may be due to incorporation of biohydrogenated FA into microbial membranes or adherence to feed particles that are easier to emulsify. Thus, if 18:1 *cis*-9 increases disruption of the FA prill in the rumen, it may provide similar benefits or may simply increase the surface area of the prill entering the abomasum and small intestine, but this requires additional investigation. Maintaining DMI and FA digestibility while increasing dietary fat allowed robust increases in FA absorption of 332 g/d of total FA and 115 and 210 g/d of 16 C and 18 C, respectively (Table 4; $P < 0.001$) and may explain the larger increase in total and preformed milk FA yield.

Transfer Efficiency of Dietary FA to Milk Fat

Transfer of consumed 18 C FA to milk fat calculated based on FA intake was 69% in control and declined to 53% with fat supplementation in the current experiment ($P = 0.01$; Table 4). Since there was no effect on FA digestibility, transfer based on observed FA digestion similarly decreased from 110% in the control to 82% in the high-fat diet.

Transfer of dietary FA to milk fat has been investigated through regression approaches and abomasal infusion of n-6 and n-3 FA that do not require tracers to quantify transfer. In a meta-regression of oil feeding, Glasser et al. (2008) reported an average of 52% recovery of dietary 18 C FA in milk fat assuming negligible contribution from extramammary tissues (positive energy balance). In abomasal infusion studies, recovery of 18:3n-3 in milk ranged from 40% to 56% with postruminal infusion of oils rich in 18:3n-3 (e.g., Hagemeister et al., 1991; Moallem et al., 2012; Yang et al., 2012).

Transfer efficiencies >100% indicate supply from non-mammary tissues either through loss of body stores or de novo synthesis. The transfer efficiency of consumed 18 C FA is often greater than 100% when dietary 18 C FA decline below 1% of diet DM (Khiaosa-ard et al., 2015). The short experimental periods of the current experiment precluded observation of changes in BW, but the cows were postpeak. Using the NASEM (2021) model, intake minus maintenance and milk production balance was -0.28 and 2.9 Mcal/d of ME and -0.19 and 1.91 Mcal/d of NEL in Fat and CON, respectively. This indicates likely near neutral energy balance in the CON. Although the mammary gland has little to no capacity for synthesis of 18 C FA by elongation of 16 C FA, elongase enzymes in other tissues may result in de novo synthesis of 18:0 and 18:1 *cis*-9, but the rate of these pathways is not well described in the cow. The 110% transfer efficiency in the control in the current experiment likely is due to synthesis of 18 C FA in adipose tissue that were partitioned to the mammary gland. Increased lipogenesis in nonmammary tissues to increase supply of 18 C FA to the mammary gland may be a homeorhetic adaptation to support milk fat yield, but the specific signaling mechanism requires additional investigation.

Kinetics of 18 C Transfer to Milk Fat Based on the Perturbation Model

A main objective of this study was to investigate the effect of dietary fat level on transfer kinetics of preformed FA to milk fat. A perturbation model using 18 C n-3 FA was selected because enriched labeled 16:0 and 18:0 are expensive. Cows have limited lipase activity, so the n-3 oil supplement was infused as free FA to aid absorption.

Urrutia et al. (2023) showed that transfer kinetics of 18:3n-3 differed from 20:5n-3 and 22:6n-3. The very-long-chain n-3 FA (20:5n-3 and 22:6n-3) have a lower total transfer to milk fat and much lower direct transfer during the first 24 h because they are rapidly moved to plasma phospholipids and cholesterol esters that are not available to the mammary gland (Palmquist, 2009). Thus, they were not selected for the current investigation because the goal was to understand kinetics of FA that are efficiently used for milk fat synthesis. Transfer of abomasally infused 18:4n-3 to milk fat was previously investigated in dairy cows by Bernal-Santos et al. (2010) and the 39% transfer was only slightly lower than the ~50% commonly reported for other 18 C FA (Moallem, 2018). The current experiment used a blend of 18:3n-3 and 18:4n-3 to allow observation of transfer of 2 different 18 C FA.

Total transfer efficiency of bolused ALA to milk fat was 51.9% in the low-fat control and was decreased 8.5 percentage units (16.4% decrease) by the high-fat diet ($P = 0.01$; Table 5). Total transfer of bolused SDA was 27.9% in the control and was decreased 5.2 percentage units (18.6% decrease) in the high-fat diet ($P = 0.01$). Transfer of SDA was lower than the 39% reported by Bernal-Santos et al. (2010) after 7 d of abomasal infusion of 57 g/d SDA rich soybean oil. Urrutia et al. (2023) had previously reported no enrichment of 20:5n-3 and 22:6n-3 after ALA infusion, but 20:5n-3 was increased 0.21 to 0.35 g of FA at peak enrichment in the current experiment (Figure 1). Stearidonic acid is more efficiently elongated and desaturated than ALA (El-Zenary et al., 2023) and utilization to synthesize downstream n-3 FA may explain the lower transfer efficiency of SDA (Bernal-Santos et al., 2010). Total n-3 transfer accounted for transfer of the elongated products to milk and was 37.6% in CON and was decreased 5.4 percentage units (14.4%) in the Fat treatment ($P < 0.01$; Table 5). However, the elongated metabolites (20:5n-3 and 22:6n-3) have a low transfer to milk fat and were likely less available to the mammary gland.

The perturbation model also allows observation of transfer through “fast” and “slow” pools. There was a treatment by time interaction on the rate of transfer of the bolused ALA, SDA, and total n-3 FA into milk fat ($P > 0.05$; Figure 2). Maximal transfer occurred over the second milking interval after the bolus when transfer of ALA, SDA, and total n-3 FA were 2.5% and 2.0%, 1.5%, and 1.2%/h in control and 1.2%, 1.75%, and 1.4%/h in Fat. The fast transfer pool has been observed to be the first 24 h and represents direct transfer from absorption and the slow pool after 24 h represents indirect transfer through tissue recycling. On average, 81% of the ALA transferred to milk occurred during the first 24 h and the remaining 18% was transferred from 24 to 146 h, and

Table 5. Transfer of an abomasal bolus of α -linolenic acid (ALA) and stearidonic acid (SDA) to milk fat and the proportion transferred through direct (first 24 h) and indirect pools (24 to 146 h)

Item	Treatment ¹		SEM	<i>P</i> -value ²
	CON	Fat		
Transferred, ³ % infused				
ALA	51.9	43.3	3.66	0.01
SDA	27.9	22.7	2.44	0.01
Total n-3	37.6	32.2	1.97	0.01
By pool, ⁴ % FA transferred				
ALA				
Direct	81.2	81.6	0.02	0.87
Indirect	18.8	18.4	0.02	0.87
SDA				
Direct	91.1	88.7	1.17	0.06
Indirect	8.87	11.3	1.16	0.06
Total n-3				
Direct	69.9	67.0	3.51	0.56
Indirect	30.1	32.9	3.52	0.56

¹Dietary treatments were a low-fat basal diet (CON) and a high-fat diet with a FA supplement that contained a blend of stearic and palmitic acid at 2% diet DM (Fat).

²Effect of treatment.

³Transfer of ALA (18:3n-3), SDA (18:4n-3), and total n-3 (sum of ALA, SDA, and 20:5n-3) FA following abomasal infusion of a free FA mixture that delivered 44 g of ALA and 20 g of SDA.

⁴Direct transfer is the rapid uptake of FA from plasma over the first 24 h and indirect transfer between 24 and 74 h from tissue recycling.

there was no effect of increasing dietary fat on the proportion through the fast and slow pools ($P = 0.87$). The direct transfer of absorbed ALA into milk fat observed in this study was similar to the 83.9% reported for mid-lactation cows by Urrutia et al. (2023). Bolused SDA was similarly primarily transferred through the fast pool, but direct transfer tended to be decreased by the high-fat diet (91.1% vs. 88.7%).

Overall, although increasing dietary fat decreased transfer of the bolused FA but had little effect on proportion through the fast direct and slow indirect mechanisms, indicating that higher transfer is likely not due to increased capacity for extraction of FA from plasma. The magnitude of this decrease was similar to the decrease in the transfer of dietary and absorbed 18 C FA to milk fat and further supports homeorhetic regulation directing FA toward milk fat synthesis when FA supply is low. Similarly, Urrutia et al. (2023) investigated transfer of ALA, eicosapentaenoic acid, and DHA to milk fat at varying stages of lactation and observed the highest transfer at peak lactation and lowest in early lactation when mobilization of endogenous FA stores is high.

CONCLUSIONS

Increasing dietary fat decreased the transfer efficiency of absorbed FA and transfer of an n-3 FA bolus to milk fat although it still increased milk fat through an increase in preformed FA. However, the proportion of FA rapidly transferred over the first 24 h did not change and indi-

cates equal downregulation of both mammary uptake of triglycerides and tissue recycling of FA. The specific metabolic adaptations supporting milk fat yield when feeding lower fat diets requires further exploration.

NOTES

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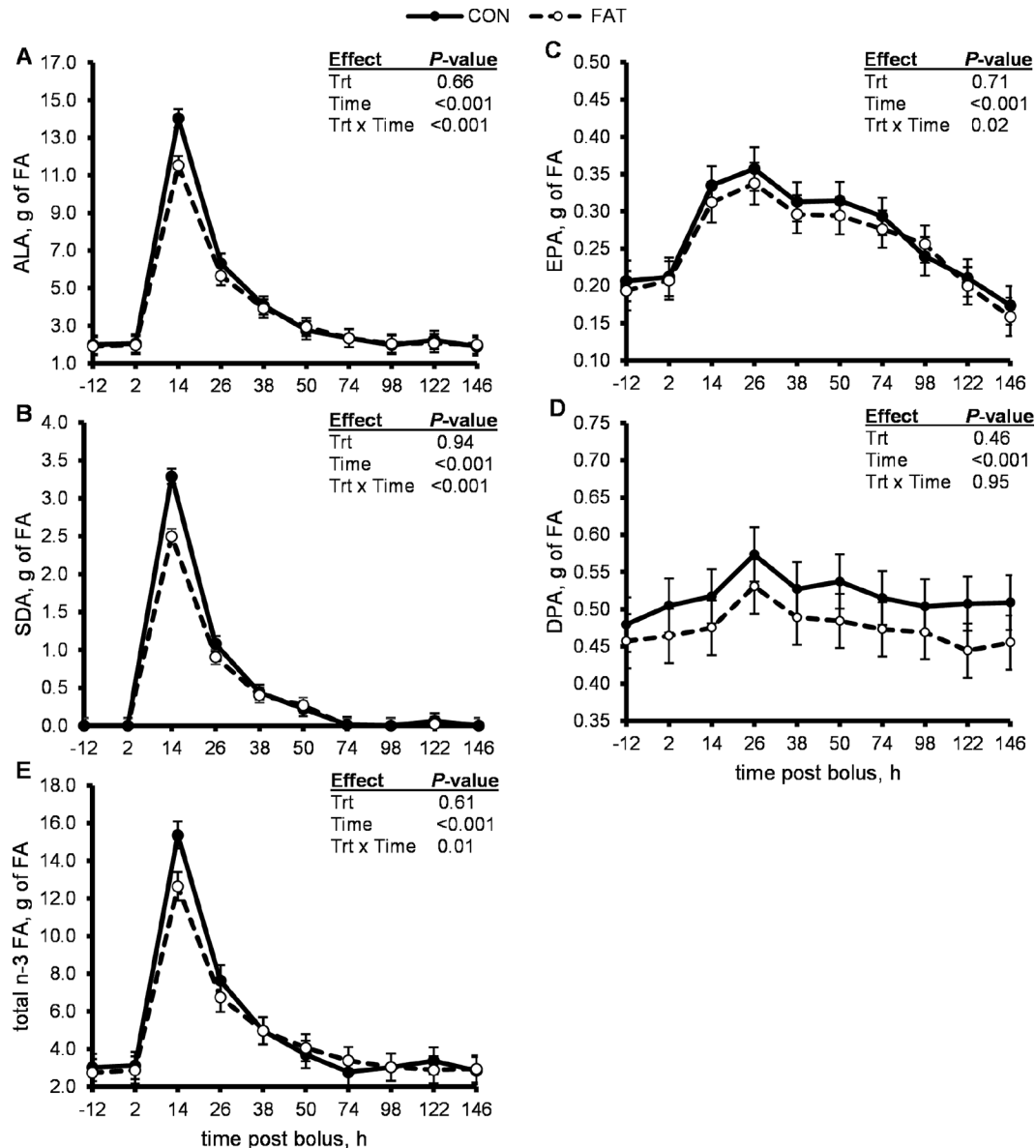


Figure 1. Effect of a blend of palmitic and stearic acid on the transfer n-3 FA from an abomasal bolus into milk fat in mid-lactation cows compared with a low-fat control. The yield of α -linolenic acid: 18:3n-3 (ALA; A), stearidonic acid: 18:4n-3 (SDA; B), eicosapentaenoic acid: 20:5n-3 (EPA; C), docosapentaenoic acid: 22:5n-3 (DPA; D), and total n-3 FA (E) in milk fat following abomasal infusion of 44 g of 18:3n-3 and 20 g of 18:4n-3 is shown relative to the bolus. Cows were either fed a low-fat basal diet alone (CON, solid line) or a high-fat diet that contained a mixture of palmitic and stearic acid at 2% DM (Fat, dashed line). Standard errors of the means are shown, and differences between treatments at each time point are shown (* $P < 0.05$). Trt = treatment.

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Nonstandard abbreviations used: ALA = α -linolenic acid; CON = low-fat basal diet; DPA = docosapentaenoic acid; EPA = eicosapentaenoic acid; FA = fatty acid; Fat = basal diet supplemented with a blend of palmitic and stearic FA fed at 2% DM; iNDF = indigestible NDF; SDA = stearidonic acid; TT = total tract.

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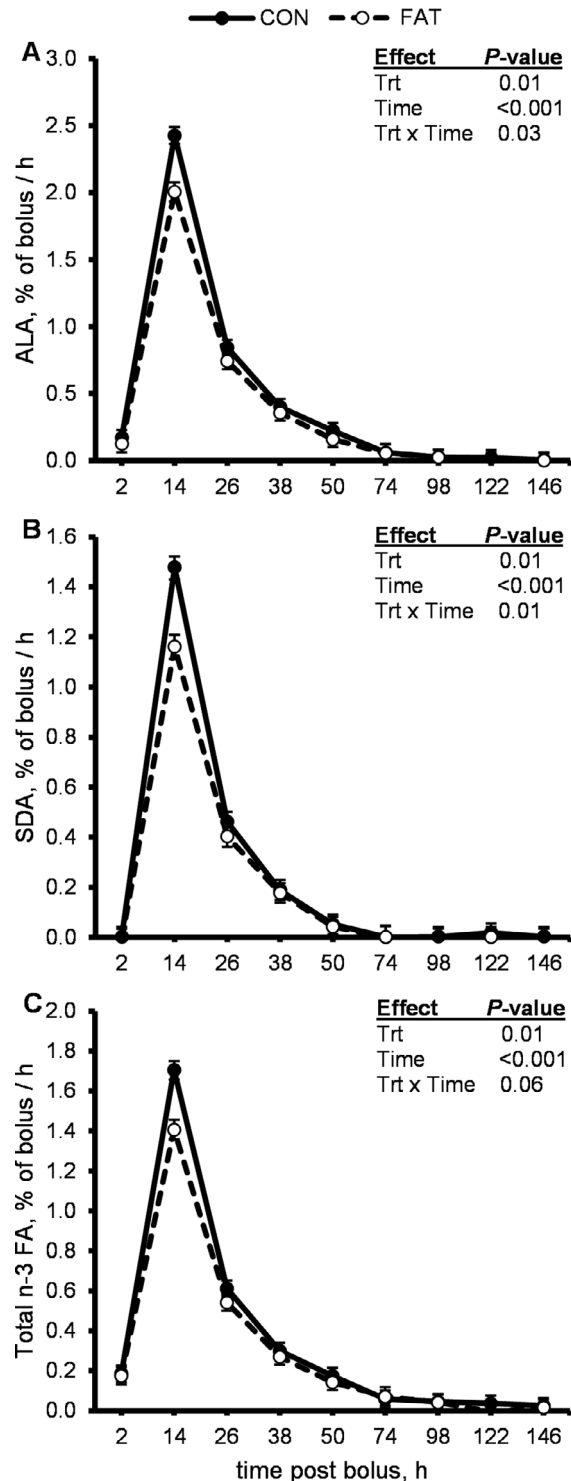


Figure 2. Effect of a blend of palmitic and stearic acid on the rate of transfer of a n-3 FA bolus into milk fat. The percentage of total transferred to milk per hour for α -linolenic acid: 18:3n-3 (ALA; A), stearidonic acid: 18:4n-3 (SDA; B), and total n-3 FA (C) following abomasal infusion of a bolus that delivered 44 g of 18:3n-3 and 20 g of 18:4n-3 is shown. Cows were either fed a low-fat basal diet alone (CON, continuous line) or a high-fat diet that fed a FA mixture of palmitic and stearic acid at 2% DM (Fat, dashed line). Means and standard error of the mean at each milking is shown. Trt = treatment.

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